GRPO for Countdown Math Problems — Project Report

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Codebase: https://github.com/sunnnybala/GStar-Assignment-2-GRPO

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1. Overview

This project implements **Group-Regularized Policy Optimization (GRPO)** to fine-tune a large language model (LLM) for the **Countdown Math Problem**, a task where the model must generate an equation using given integers exactly once to reach a specified target.

The work focused on stable reinforcement learning for symbolic reasoning, emphasizing **safe equation evaluation**, **precise reward design**, and **robust on-policy PPO training**.

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2. Methodology

Task Definition:
Given a list of integers `[n₁ … n ]` and a target `T`, the model must output a valid equation that evaluates exactly to `T`, using each number once.

Dataset:
`justinphan3110/Countdown-Tasks-3to4` with train/test splits.
Prompt format encourages structured reasoning within `<think>...</think>` and final output in `<answer>...</answer>`.

Base Model and Setup:

- **Base LLM:** *Qwen/Qwen3-1.7B* (bf16 precision, FlashAttention-2).
- **Inference Engine:** *vLLM 0.7.2* with prefix caching.
- **Training Framework:** *Modal* for reproducible cloud execution and persisted volumes.

Reward Design:
Three-tiered, sequence-level reward:

Reward	Condition
1.0	Valid equation using the exact multiset of provided numbers and reaching the target
0.1	Equation present but incorrect or with wrong numbers
0.0	No valid equation detected

The reward function includes:

- Multiset-based number validation via `Counter`.
- A **safe mathematical evaluator** using a shunting-yard parser with float tolerance (|result−target| < 1e-3).

GRPO Advantage Normalization:

- Each prompt has $G = 8$ sampled responses.
- Rewards are normalized per group
- This emphasizes *relative quality* within each group and stabilizes updates.

Masked PPO Loss:

- Token-wise clipped surrogate: $L=-\min(A\cdot r, A\cdot \text{clip}(r, 1-\varepsilon, 1+\varepsilon))$
- Reductions averaged **per sequence, then across batch** for fair scaling independent of response length.

Hyperparameters:

Parameter	Value
Steps	80
Rollout batch	128
Group size	8

Grad accum.	32
LR	7e-6
Optimizer	AdamW (β=(0.9, 0.95))
Clip range	0.2
Weight decay	1e-2

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3. Experiments and Evaluation

Training Protocol:
Each GRPO step performs a full rollout via vLLM, computes rewards and normalized advantages, then optimizes with PPO under masking corrections. Evaluation occurs every 10 steps on the full test set.

Metrics:

- **Accuracy:** fraction of samples with reward = 1.0
- **Mean reward, Std reward**
- **Average output tokens**
- **Counts:** correct / partial / failed

Results (Step 80) (Max Token =256):

Metric	Value
Accuracy	**48.7 %**
Mean Reward	**0.487**
Avg Tokens	230.8

Results (Step 80) (Max Token =512):

Metric	Value

Accuracy	**59.2 %**
Mean Reward	**0.592**
Avg Tokens	502.8

Experiments and Observations:

- Tried multiple permutations of the reward function
- Proper masking eliminated bias toward longer outputs.
- I tried running two phase training where the model focussed on formatting first and then result but i didnt get good results. The model collapsed in first phase.
- Further improvement possible via KL-controlled training (eleiminating policy drift).

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4. Infrastructure and Reproducibility

- **Environment:** CUDA 12.4.1 + Torch 2.4.1 + vLLM 0.7.2 + Flash-Attn 2.7.4.post1
- **Logging:** TensorBoard events under `/root/output/tb/hw_a2_<loss_type>/`
- **Artifacts:** Models and tokenizer persisted to `/root/output/hw_a2_solution_<timestamp>`

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5. Ongoing Work (Problem 2): GRPO for Drug Discovery

A parallel ongoing project applies the **GRPO framework to drug discovery**, where the **reward model** is based on *predicted drug efficiency and potency scores*.
The aim is to train generative policies that propose molecular structures with high therapeutic

potential while maintaining chemical validity.
This work is still in progress and not yet public.

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6. Conclusion

The GRPO implementation achieved strong performance on symbolic reasoning tasks like Countdown, demonstrating that **group-normalized PPO with safe evaluation** can effectively train small LLMs for structured arithmetic generation.
Future extensions include **KL-controlled PPO**, **multi-epoch updates**, and applying the same framework to **drug design reward modeling** for real-world scientific generation tasks.